James Bray

R&D CONTRACTOR

Aerospace MEng graduate specialising in UAV system software, with a broad base of projects across firmware, controls and simulation. I drop into cross-disciplinary teams, move fast between system design and hands-on build, and ship working results.

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Experience

Mission Software Engineer (Contract) — Windracers JUL 2022 - OCT 2022

- Integrated features into the Mission Software using the C#/.NET with Angular
- Built custom STM32 development environments, migrating the team off outdated compilers
- Operationalised a ML health monitor to detect and diagnose engine or sensor faults
- Deployed custom AI microservices on AWS, including an automated ATC chatbot
- Completed simulator and flight training on RC planes used for comms equipment testing

Integration Engineer (Contract) — XPRIZE AURA SEP 2025 - JUN 2026

- Deployed in Alaska to develop a swarm of autonomous UAVs for wildfire detection and mitigation
- Engineered a novel thermal detection refinement method to localise fires with 1 m accuracy
- Profiled the system architecture to identify and mitigate critical network bottlenecks
- Ran rapid hardware-in-the-loop testing cycles across a 38 h non-stop development sprint

Propulsion Avionics Engineer (University Team) — Sunride SEP 2024 - MAY 2026

- Designed avionic systems for the university's first liquid rocket engine test stand
- Used LabJack DAQ systems to interface sensors and actuators for precise control
- Built a Simulink fluid model with PID control to regulate fuel and oxidiser flow
- Implemented fault detection and safety protocols for engine test procedures

Assistant Project Engineer (Internship) — AMRC JUL 2024 - OCT 2024

- Designed and manufactured a £4,000 partial-discharge testing cell for 20 kV motor coils
- Coordinated with 6 suppliers to source components for manufacturing
- Managed deliverables, mitigated risks and produced supporting documentation
- Used Ansys FEA to evaluate geometries for slinky-stator manufacture and a robot end effector

Education

Aerospace Engineering MEng — University of Sheffield SEP 2022 - MAY 2026

Masters, first-class. Avionics-focused: electronic systems and control theory applied to aerospace.

- Dissertation: 'Risk-Aware Multi-UAV Search Strategies for Large-Scale Wildfire Detection' (86%)
- Favourite modules: State-Space Control Design (92%), Multisensor Decision Systems (87%), Aircraft Dynamics & Control (85%), Real-Time Embedded Systems (79%), Data Modelling and Machine Intelligence (76%)

A-Levels & GCSEs — George Spencer Academy SEP 2015 - JUL 2022

- A-Levels: A*AAA in Maths, Further Maths, Computer Science, Physics
- GCSEs: 4 Grade 9s (Maths, Biology, Chemistry, Physics); grade 7–8 in Computer Science, D&T, Geography & English

Proficiencies

TECHNICAL

- Avionics & control systems
- Algorithm development
- Embedded / real-time firmware
- Advanced manufacturing

SOFTWARE

- Python, C, C#
- MATLAB & Simulink
- Git & version control
- Unity, Blender

STRENGTHS

- Cross-disciplinary R&D
- System optimisation
- Analytical problem-solving
- Leadership & collaboration